

Air-Raid Alerts in Ukraine

Patterns in 101,705 alert records across 25 regions, 25 February 2022 – 23 June 2026.

For more than four years, air-raid alerts have been part of daily life across Ukraine. Each one is a moment recorded with a time, a place, and an end. Put together, those moments form a long, detailed record — and that record has a shape. This report reads that shape: **when** alerts happen, **where** they concentrate, whether the rhythm is seasonal or simply rising, and where the recent trend points.

Disclaimer. This document is a **data analysis only**. It describes statistical patterns in a public dataset of air-raid alert records. It does **not** contain, and should not be read as, political or military analysis of the war — no assessment of causes, strategy, responsibility, or outcomes is offered or implied.

101,705

ALERTS RECORDED

25

REGIONS (OBLASTS)

May 2026

BUSIEST MONTH
(3,402)

~136/day

PROJECTED END-2026

The shape of four years

The clearest way to begin is simply to count alerts month by month. The result is not a steady line. After the spike of the invasion in early 2022, alerts settled

into a calmer 2023, climbed gradually through 2024, dipped in mid-2025 — and then rose sharply.

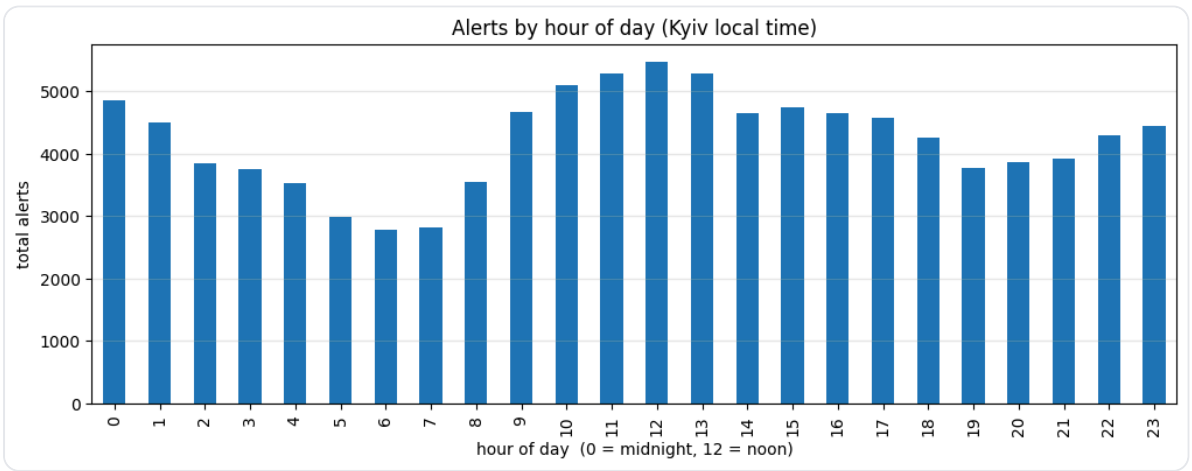


Air-raid alerts per month, all Ukraine. The final month is partial.

The most intense period of the war is the present. The single busiest month on record is **May 2026, with 3,402 alerts** — higher than the February–March 2022 invasion peak (2,084/month).

When in the day, when in the week

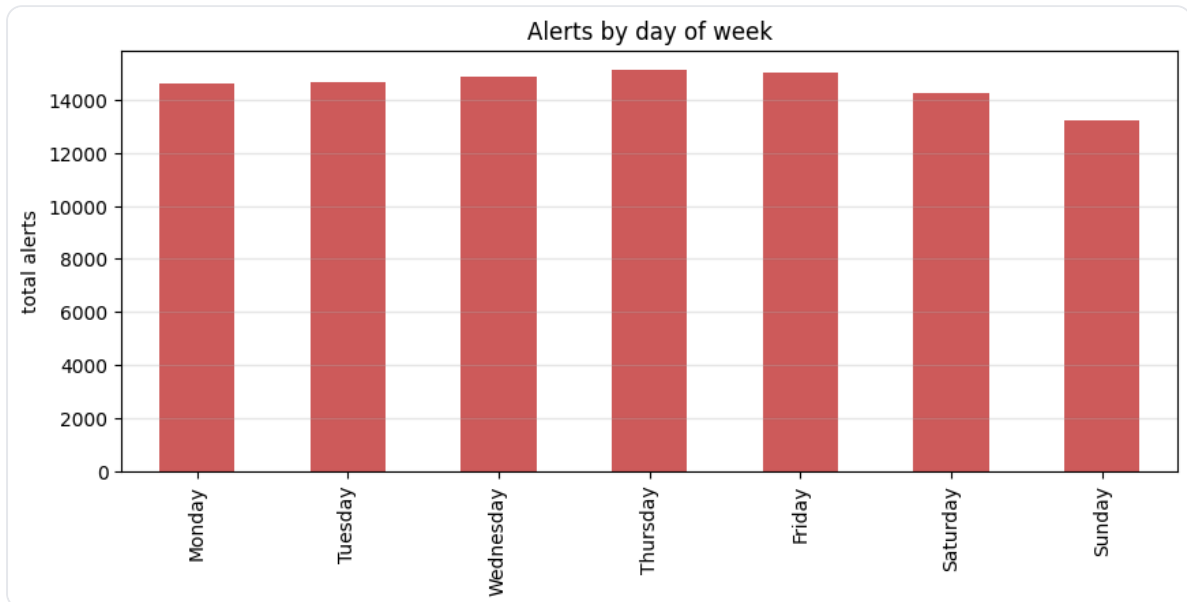
Zooming from years down to a single day, a rhythm appears. Counting when alerts *begin* (in Kyiv local time), activity is lowest in the small hours of the morning and peaks around midday.



When alerts start, by hour of day (Kyiv local time).

One honest caveat matters here: this counts when an alert *starts*, not how long the country spends under alert. Overnight drone alerts tend to be long, single events — so the hours actually spent sheltering likely skew more nocturnal than the midday peak suggests.

Across the week, there is almost no pattern at all:

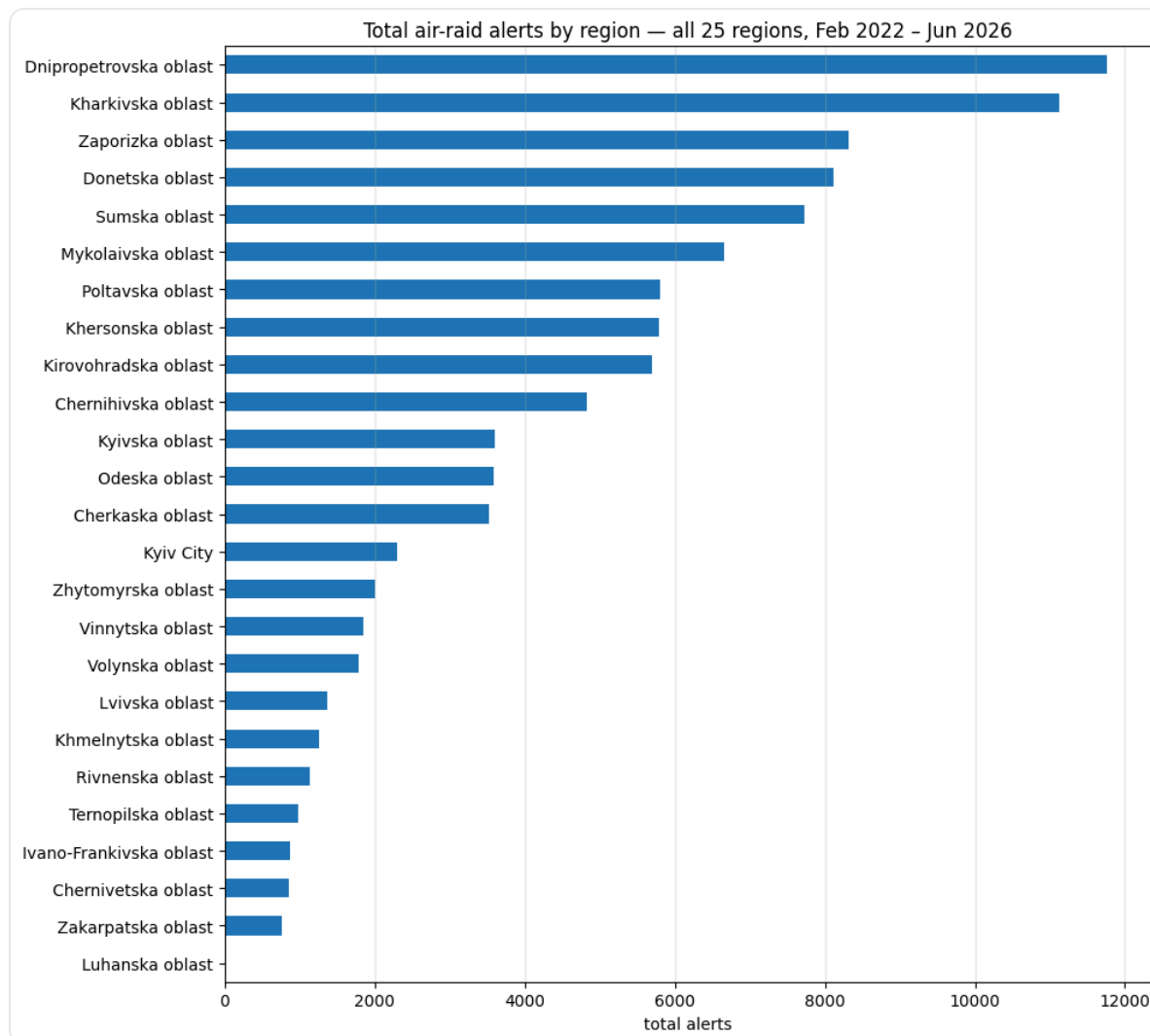


Total alerts by day of week.

The war keeps no weekly calendar. Thursday is the busiest day and Sunday the quietest, but Thursday sits only about **14% above** the quietest day. Attacks do not pause for weekends.

Where the alerts fall

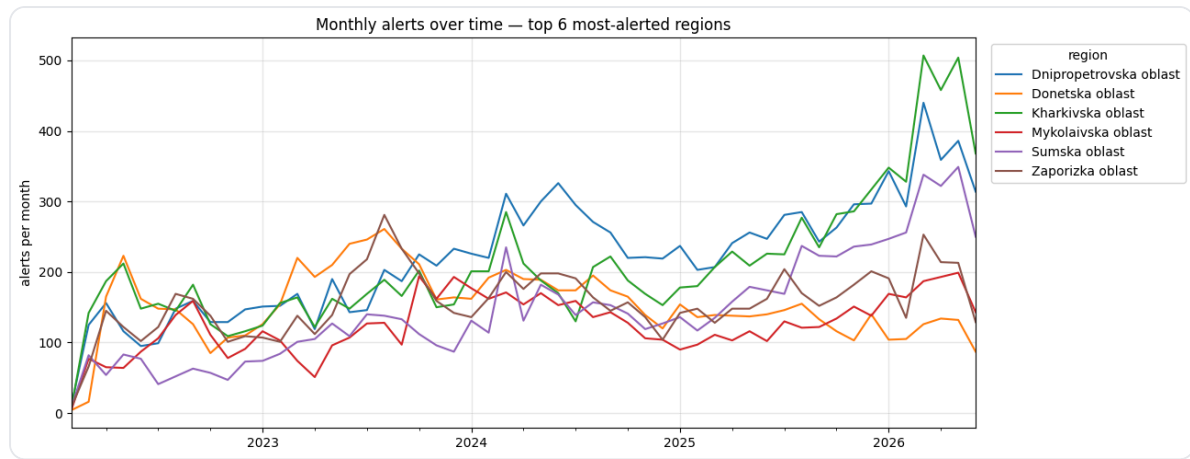
Geography is far less even than the calendar. A handful of frontline regions in the east and south absorb most of the alerts.



Total alerts by region, all 25 oblasts.

The five most-alerted regions are all on the front line: Dnipropetrovsk (11.6%), Kharkiv (10.9%), Zaporizhzhia (8.2%), Donetsk (8.0%) and Sumy (7.6%). Notably, the capital — **Kyiv City** — **sits far down the list**, with 2,299 alerts.

That map has not stayed still. Tracking the most-alerted regions over time shows them rising and falling together in broad national waves — but pulling apart sharply in 2025–2026.

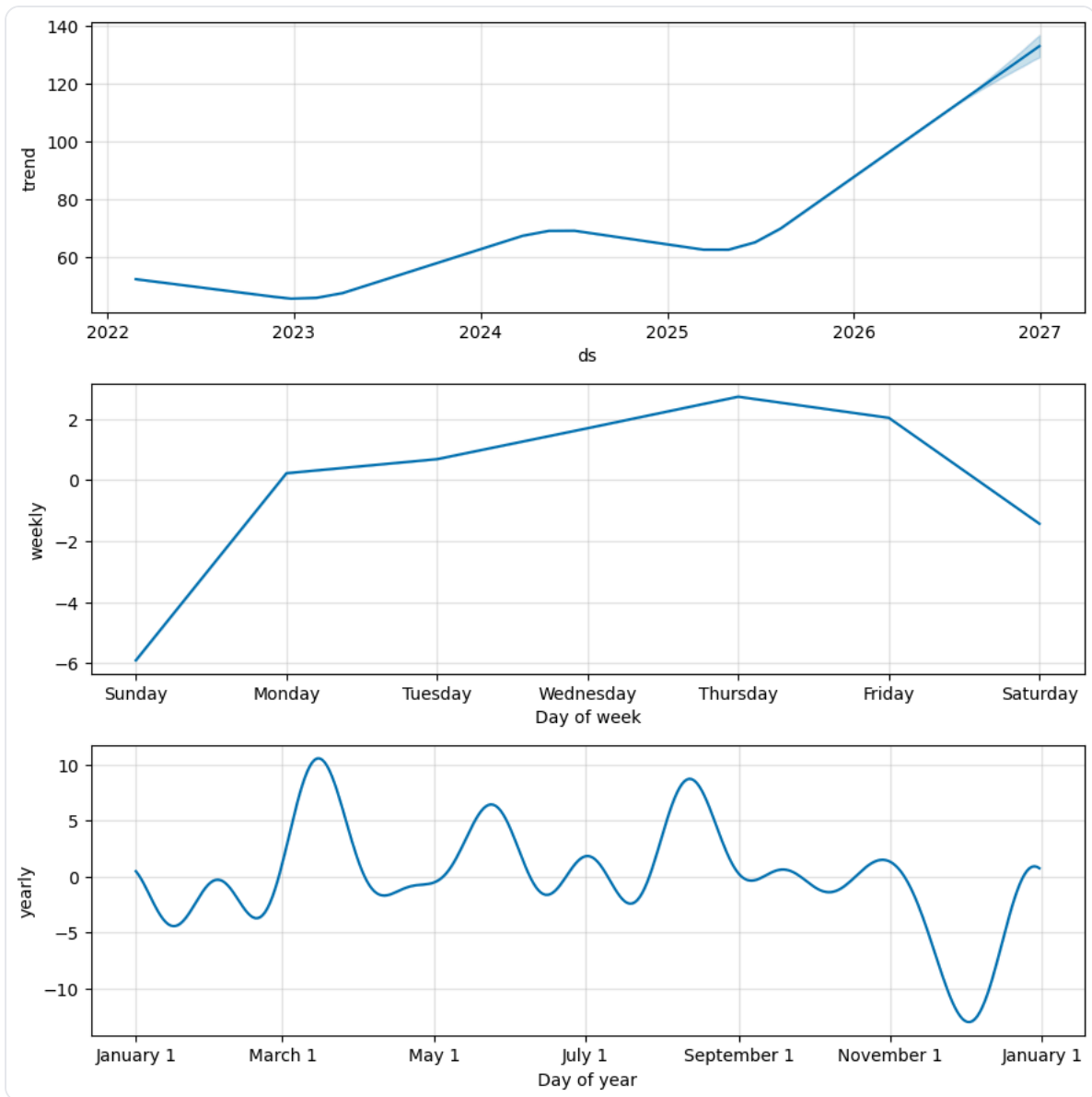


Monthly alerts over time for the six most-alerted regions.

Kharkiv and Sumy, on the northeastern border, climb to the top of the chart, while Donetsk — the leader in 2023 — falls back. (One caveat: part of Donetsk's decline may reflect occupied areas no longer issuing alerts, rather than a genuine reduction in danger.)

A rising trend, not a season

A natural question is whether alerts follow a yearly season — heavier in winter, lighter in summer. To test this properly, the daily series was decomposed into its underlying components: a long-run trend, a weekly cycle, and a yearly cycle.



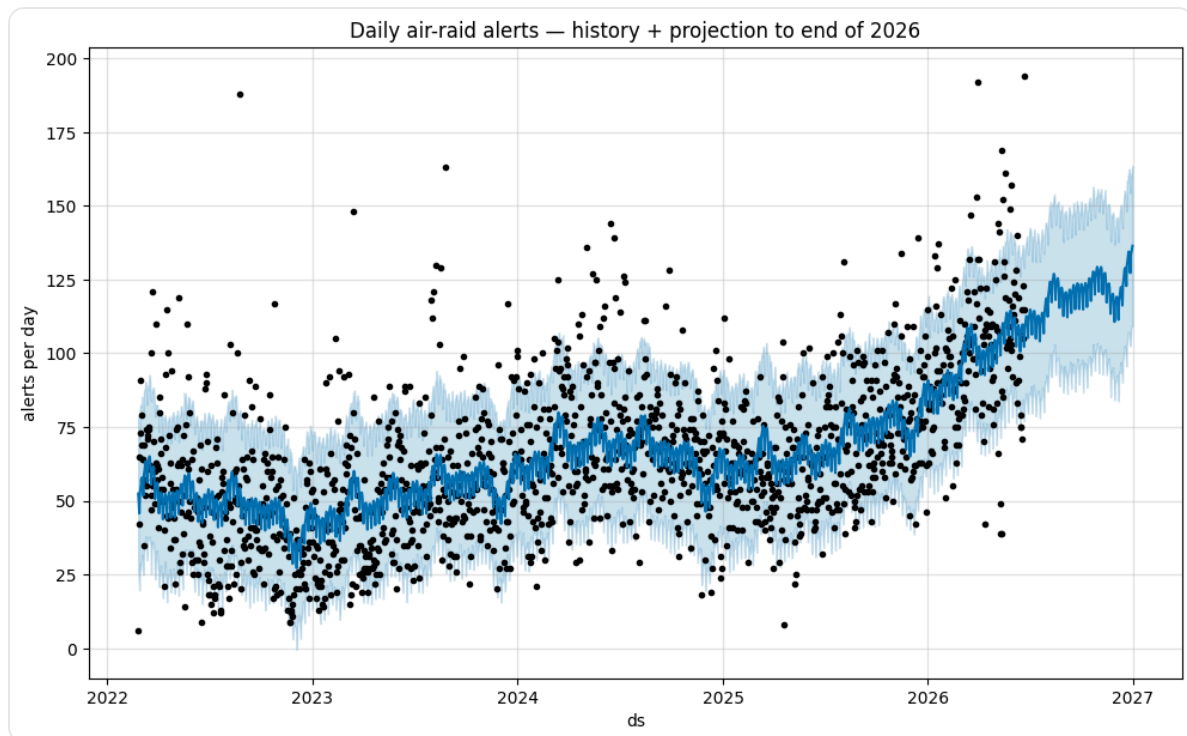
The series split into long-run trend (top) and weekly cycle (bottom).

The dominant signal is a trend, not a season. The long-run trend swings from about 46 to 133 alerts per day, dwarfing the weekly cycle (a few alerts) and the yearly one (about a dozen). The apparent yearly wobble is small and statistically unreliable on only four years of data — so no real calendar season should be read into it.

Where the trend points

Finally, extending the trend forward gives a sense of direction. If the recent escalation continues at its current pace, the model projects alert volume to

keep climbing through the end of 2026.



Daily alerts (points) with model fit and projection to 31 December 2026 (shaded band = uncertainty).

Projection: roughly 136 alerts per day by the end of 2026 (uncertainty range 109–164) — sustained record highs. **This is an extrapolation, not a prediction.** Air-raid alerts are driven by military and strategic decisions, not a natural process; a ceasefire, an escalation, or a change in air defence would break the projection entirely. The widening band reflects that growing uncertainty.

Data, method & limitations

- **Source.** A public dataset of air-raid siren records (the Vadimkin *ukrainian-air-raid-sirens-dataset*, MIT-licensed), originating from the volunteer *eTryvoga* channel. This is volunteer / open-source data, not official government data.
- **Reproducibility.** The exact data snapshot (frozen 23 June 2026) and the full analysis notebook are published openly, so anyone can re-run and check

every figure.

- **Method.** Timestamps were converted from UTC to Kyiv local time (handling the daylight-saving switch), counts were aggregated by hour, day, month and region, and the forecast uses an additive time-series model (Prophet).
- **Limitations.** About 4.9% of records have an estimated end time; two permanently-active sirens (Luhansk, Crimea) are excluded; counts are of alert *events*, not hours spent under alert.

Data: Vadimkin/ukrainian-air-raid-sirens-dataset (MIT), via the eTryvoga volunteer channel. Analysis & charts generated from the open notebook at github.com/PavloLoiek/ukraine-air-raid-alerts.

Prepared June 2026. Code under MIT licence.

Data analysis only — not political or military analysis.